DecoR: deconfounding time series with robust regression [3]

Joon Kim

Seminar "Advances in Statistical Inferences"

May 21, 2026

Keywords: Causal Inference · Robust Regression · Time Series Analysis

Outline

1. Introduction
2. Theoretical Setting
3. Methodology
4. Theoretical Guarantees
5. Experiments
6. Conclusion & Future Work

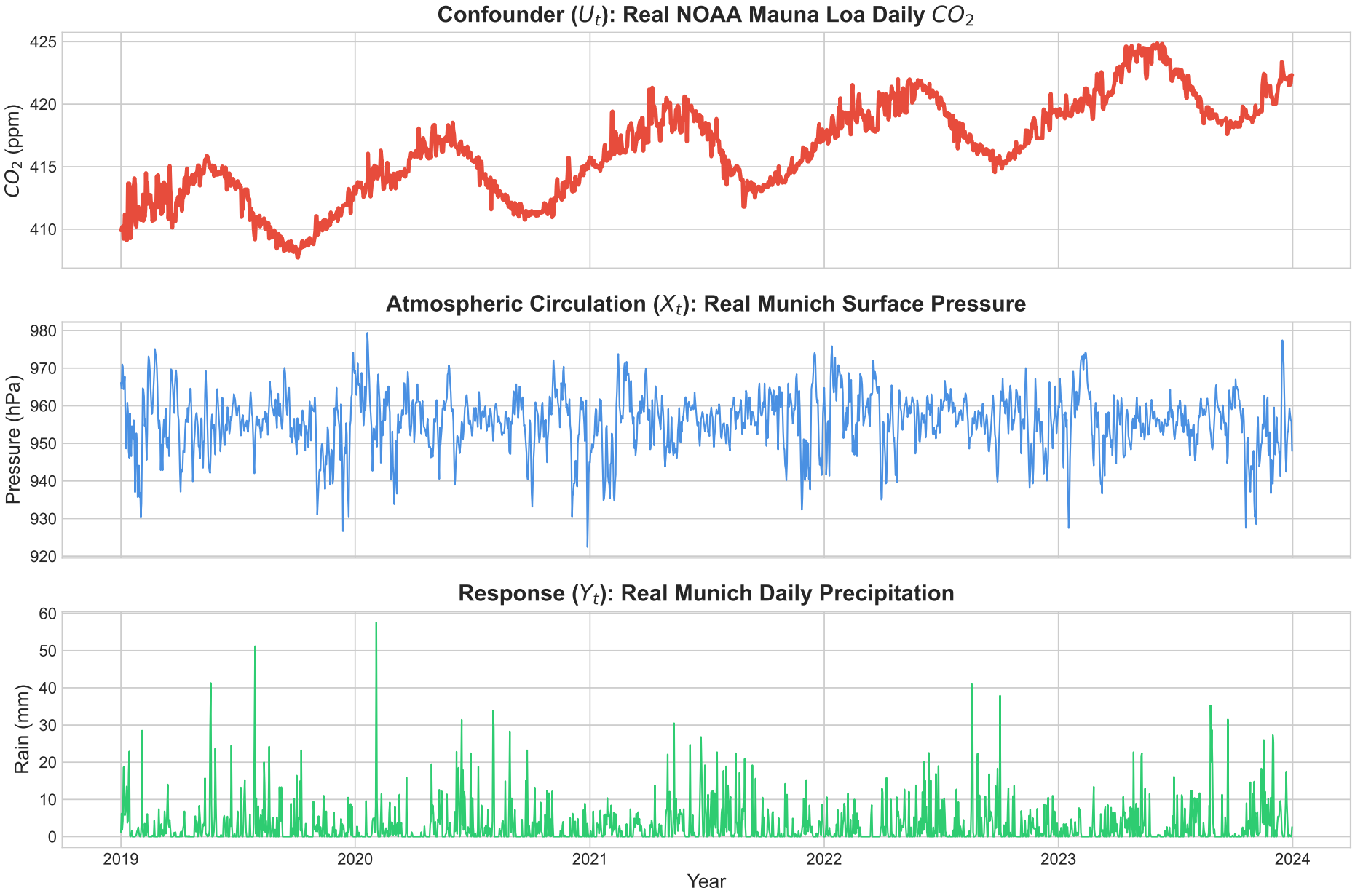


Figure: Real climate data fetched from Open Meteo [5] and Mauna Loa Observatory [2]

Importance of Decounfounding - Earth Science

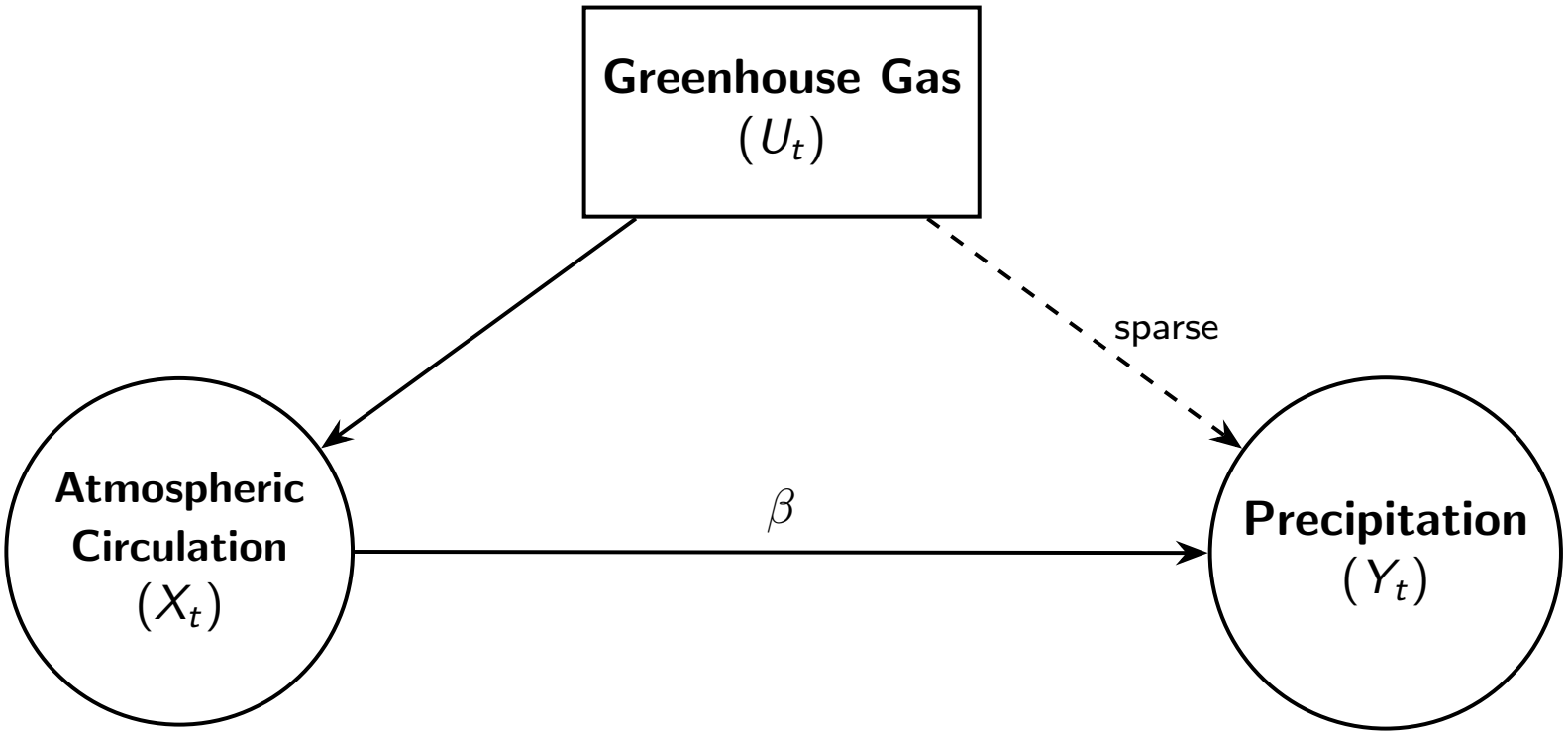


Figure: Causal Graph of Greenhouse Gas Confounding [4]

- **Goal:** Estimating β (effect of X to Y)
- **Problem:** Biased estimation of β (via standard OLS) without considering hidden confounder U
- **Idea:** Removal of confounder U seems important for accurate estimation of β

Outline

1. Introduction
2. Theoretical Setting
3. Methodology
4. Theoretical Guarantees
5. Experiments
6. Conclusion & Future Work

The Underlying Model

Setting 1: Data Generating Process

Let $T \in \mathbb{R}_{>0}$ be a fixed time horizon and $d \in \mathbb{N}$. We consider the relationship:

$$Y_t = X_t^\top \beta + U_t + \eta_t, \quad t \in [0, T] \tag{1}$$

Components:

- $X_t \in \mathbb{R}^d$: a (multivariate) stochastic process.
- $U_t \in \mathbb{R}$: a (univariate) stochastic process.
- η_t: i.i.d. centered Gaussian noise independent of X.
- $\beta \in \mathbb{R}^d$: True causal effect (estimand).

Observations: Given a fixed $n \in \mathbb{N}$, we assume that we observe with $\Delta t = T/n$:

- $X^n := (X_{\Delta t}, X_{2\Delta t}, ..., X_{n\Delta t})^\top$
- $Y^n := (Y_{\Delta t}, Y_{2\Delta t}, ..., Y_{n\Delta t})^\top$

In short: X^n and Y^n are vectors of discrete (equidistant) n observations, U_t is a hidden confounder.

Frequency Domain Transformation

Recall: Only discrete $X^n := (X_{\Delta t}, X_{2\Delta t}, ..., X_{n\Delta t})$ in time domain is given, not the continuous X_t with $t \in [0, T]$

Basis Expansions

Given only X^n (n discrete observations), "reconstruct" the original process X_t as a linear combination of basis functions:

$$\hat{X}_{i\Delta t} = \sum_{k=1}^{K} \hat{c}_k \cdot \phi_k(i\Delta t) \tag{2}$$

- We use same basis functions to "reconstruct" Y^n (and U^n)
- Here, we will mainly use the (ordered) **orthonormal** cosine basis.

Frequency Domain Transformation - 1D Example

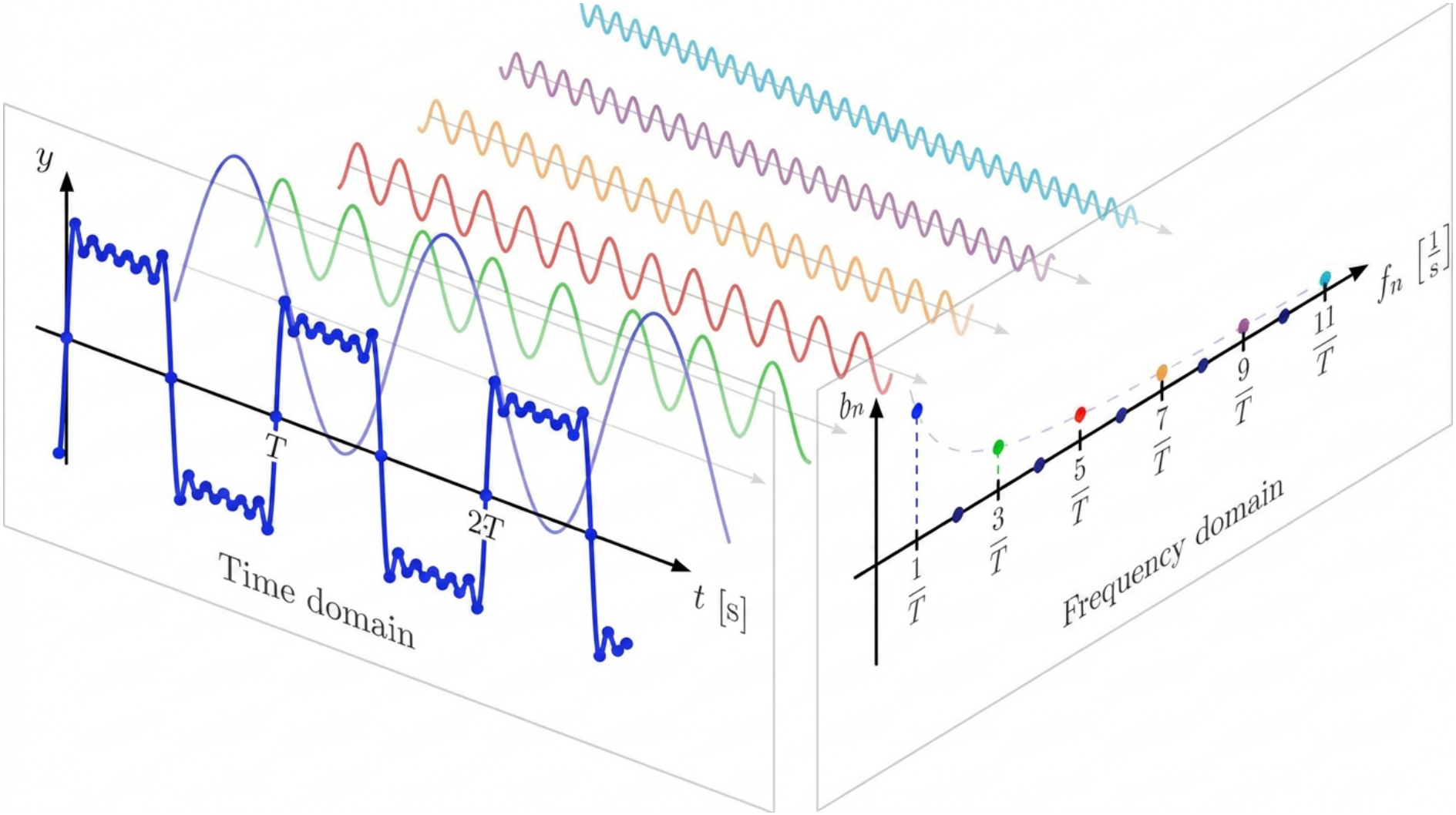


Figure: Given the discrete sample points in time domain, continuous X_t is "reconstructed" by fitting the coefficients of the basis functions.

Frequency Domain Transformation (Cosine Basis)

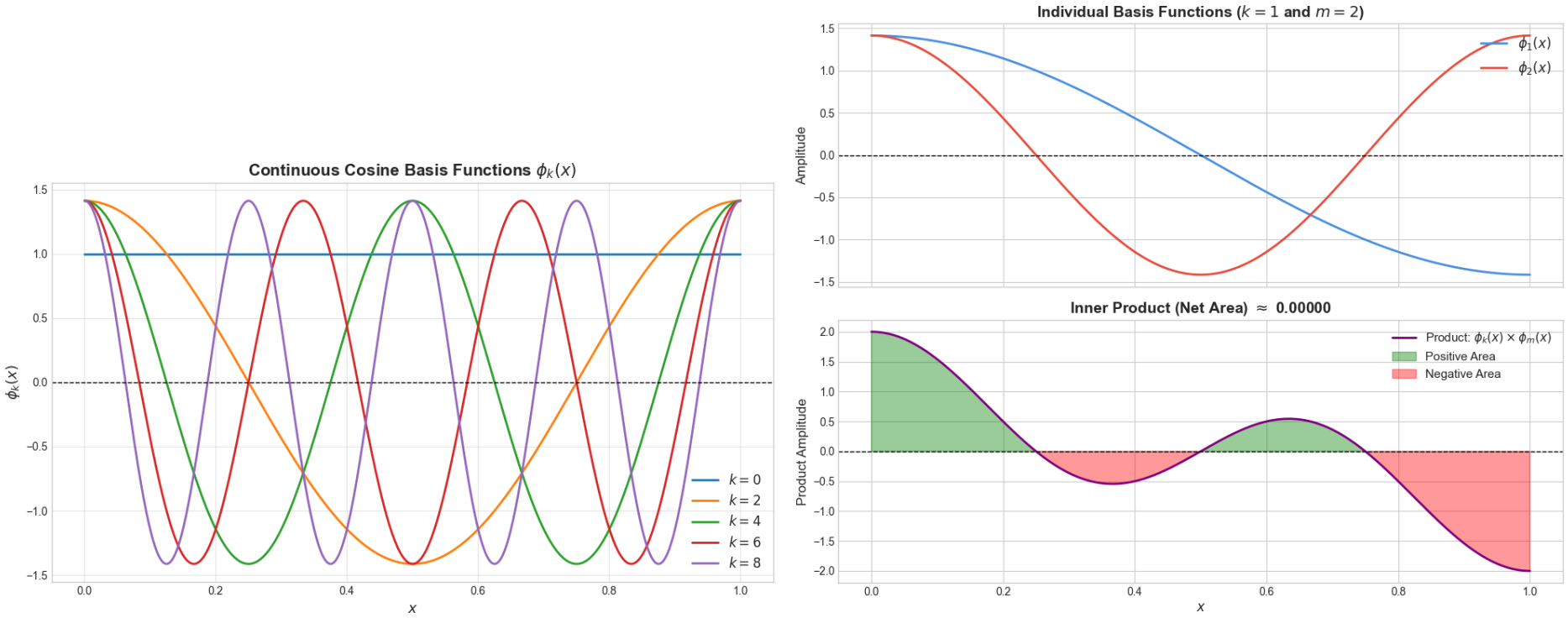


Figure: **Left:** Cosine Basis Functions.
Right: Orthogonality of Cosine Basis Functions; Inner product between basis functions of different frequencies results in zero.

Frequency Domain Transformation (Cont.)

Estimating coefficients c

let Φ be the $n \times K$ matrix with entries $\Phi_{i,k} = \phi_k(i\Delta t)$ and $K =$ number of basis functions. We want to estimate the vector of coefficients $c = (c_1, ..., c_K)$. We consider the sum of squared errors:

$$\text{SSE}_X = \sum_{i=1}^{n} (X_{i\Delta t} - \hat{X}_{i\Delta t}) = (X^n - \Phi c)^\top (X^n - \Phi c) \tag{3}$$

The least squares solution gives $\hat{c} = (\Phi^\top \Phi)^{-1} \Phi^\top X^n$.
For **orthonormal** basis, it holds that $\Phi^\top \Phi \approx n I_K$, which gives $\hat{c} \approx \frac{1}{n} \Phi^\top X^n$.

Above, we used:

- For orthonormal basis, it holds: $\int_0^T \phi_k(t)\phi_m(t)dt = \begin{cases} 1 & \text{if } k = m \\ 0 & \text{if } k \neq m \end{cases}$
- For $k = m$: Sum approximation yields: $\frac{1}{n} \sum_{i=1}^{n} \phi_k(i\Delta t)^2 \approx 1$.

Frequency Domain Transformation

Previous derivation: $\hat{c} \approx \frac{1}{n}\Phi^\top X^n$.

Transformation $T_k^{\phi,n}$

Let $\mathcal{D}$ be the set of n $\mathbb{R}^d$-valued discrete samples of stochastic processes (see X^n) and $\mathcal{C}$ the set of random variables on $\mathbb{R}^d$. Let timestamp $\Delta t = T/n$. Define, for all $k \leq n$, the function $T_k^{\phi,n} : \mathcal{D} \to \mathcal{C}$ by:

$$T_k^{\phi,n}(V) := \begin{bmatrix} \frac{1}{n}\sum_{i=1}^n (V_{i\Delta t})_1 \phi_k(i\Delta t) \\ \vdots \\ \frac{1}{n}\sum_{i=1}^n (V_{i\Delta t})_d \phi_k(i\Delta t) \end{bmatrix}$$

- $k \leq n$: Integer indices (Frequencies) of (ordered) orthonormal basis functions $\leq$ n (# data points)
- **DCT:** for the cosine basis, this corresponds to Discrete Cosine Transform

Assumptions and Sparsity

Definition 1: (ϕ, G)-sparse process

A process U is (ϕ, G)-sparse with respect to basis ϕ if for a set $G \subseteq \mathbb{N}$:

$$\langle U, \phi_k \rangle_{L^2} = \int_0^T U_t \phi_k(t) dt = 0 \quad \text{for all } k \notin G \text{ almost surely}$$

Assumption 1: Sparse Confounding

The set G is "small," meaning the hidden confounder only affects a few basis components (frequencies).

Implication:

- From before: $\int_0^T U_t \phi_k(t) dt \approx \frac{1}{n} \sum_{i=1}^n (U_{i\Delta t}) \phi_k(i\Delta t) = T_k^{\phi,n}(U)$
- By transforming the data, we can isolate the "clean" frequencies G^c where the confounder disappears.

Example: Sparse U

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.fft import dct

n = 500
T = 1.0  # Time horizon
t = np.linspace(0, T, n, endpoint=False)

# some small frequency set
k1, k2 = 2, 5

U_time = 2.0 * np.cos(2 * np.pi * k1 * t / T) + 1.0 * np.cos(2 * np
    .pi * k2 * t / T)

U_freq = dct(U_time, type=2, norm='ortho')
k_indices = np.arange(len(U_freq))
magnitude = np.abs(U_freq) / (n / 2) # Normalize

# 'magnitude' is zero everywhere except
# exactly at the integer indices k=2 and k=5

```

Example: Sparse U

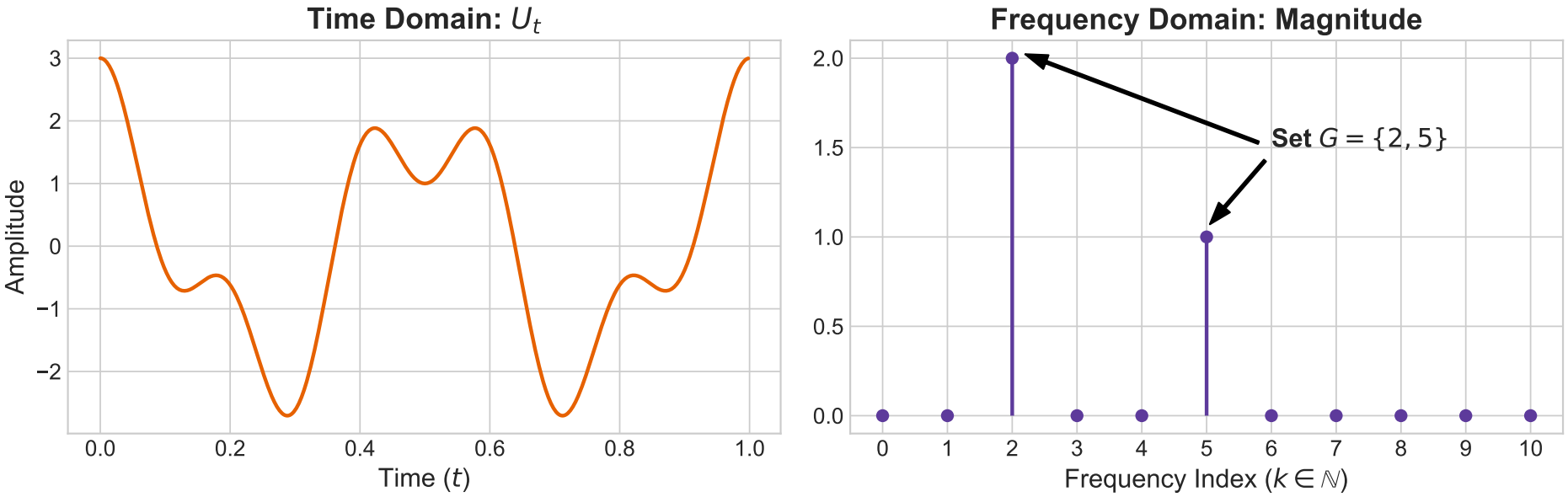


Figure: A sparse confounder U_t in the time domain (left) and frequency domain (right). The non-zero spikes represent the "small" confounded set G.

Frequency Domain Transformation - Outlier Problem

from before:

$$T_k^{\phi,n}(Y) = \begin{bmatrix} \frac{1}{n}\sum_{i=1}^{n}(Y_{i\Delta t})_1 \phi_k(i\Delta t) \\ \vdots \\ \frac{1}{n}\sum_{i=1}^{n}(Y_{i\Delta t})_d \phi_k(i\Delta t) \end{bmatrix} \quad \text{and} \quad Y_t = X_t^\top \beta + U_t + \eta_t, \ t \in [0, T].$$

The Deconfounded View

Using the linearity of $T_k^{\phi,n}$, it holds for all $k \leq n$:

$$\begin{aligned} T_k^{\phi,n}(Y) &= T_k^{\phi,n}(X)^\top \beta + T_k^{\phi,n}(U) + T_k^{\phi,n}(\eta) \\ &\overset{\text{a.s.}}{=} \begin{cases} T_k^{\phi,n}(X)^\top \beta + T_k^{\phi,n}(U) + T_k^{\phi,n}(\eta) & \text{if } k \in G, \\ T_k^{\phi,n}(X)^\top \beta + T_k^{\phi,n}(\eta) & \text{if } k \notin G. \end{cases} \end{aligned} \tag{4}$$

Remark: The Outlier Problem

If $k \in G$, $T_k^{\phi,n}(U)$ acts as an **adversarial outlier**. If $k \notin G$, the confounder term is zero, leaving an unconfounded linear relationship.

Outlier Problem

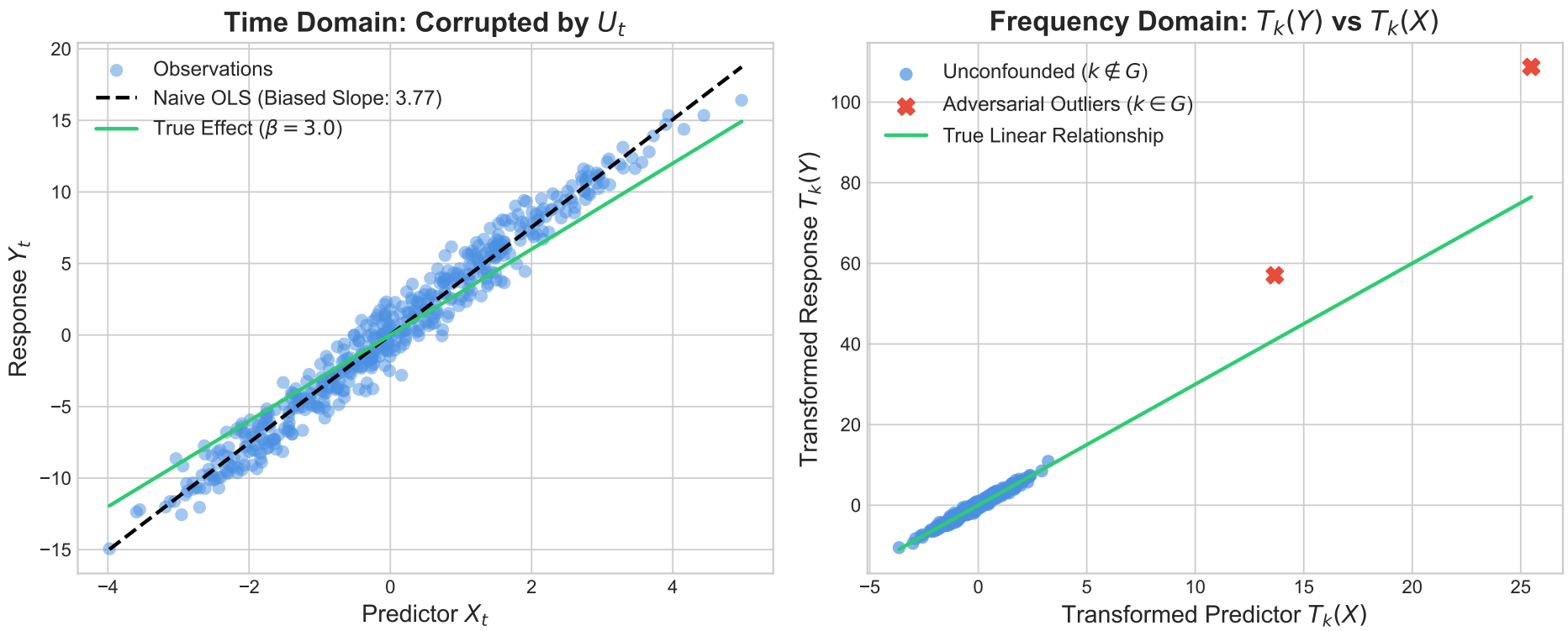


Figure: **Left:** In the time domain, the hidden confounder U_t corrupts X_t and the estimation of β via OLS. **Right:** Applying DCT $T_k^{\phi,n}$ isolates the confounder into the sparse set G. The goal becomes identifying these isolated outliers and estimating the unconfounded relationship β (via Robust Regression).

Outline

1. Introduction
2. Theoretical Setting
3. Methodology
4. Theoretical Guarantees
5. Experiments
6. Conclusion & Future Work

The DeCoR Algorithm

Algorithm 1: Deconfounding via Robust Regression

Require: $X^n \in \mathbb{R}^{n \times d}$, $Y^n \in \mathbb{R}^n$, orthonormal basis ϕ, robust linear regression algorithm $\mathcal{A}$

$X^n_\phi \leftarrow (T_1^{\phi,n}(X), \ldots, T_n^{\phi,n}(X))^\top$ ▷ Translate to frequency domain
$Y^n_\phi \leftarrow (T_1^{\phi,n}(Y), \ldots, T_n^{\phi,n}(Y))^\top$
$\hat{\beta}^{\phi,n}_{\text{DeCoR}} \leftarrow \mathcal{A}(X^n_\phi, Y^n_\phi)$ ▷ robust regression
return $\hat{\beta}^{\phi,n}_{\text{DeCoR}}$

Insights:

- **Modular Design:** Complexity shifted to the transformation step ($T^{\phi,n}$).
- **Flexible basis:** Requires only an orthonormal basis ϕ (e.g., Cosine).
- **Plug-and-play solver:** Uses any standard robust regression algorithm $\mathcal{A}$.

The Torrent Algorithm [1]

Algorithm 3: Torrent (Bhatia et al., 2015)

Require $X \in \mathbb{R}^{n \times d}$, $Y \in \mathbb{R}^n$, $a \in \{1, \ldots, n\}$

$S_0 \leftarrow \{1, \ldots, n\}, e \leftarrow Y, \text{err} \leftarrow \infty, t \leftarrow 0$

while $\|e\|_2 < \text{err}$ **do**

  $t \leftarrow t + 1$

  $\text{err} \leftarrow \|e\|_2$

  $\hat{\beta}^t_{\text{Tor}} \leftarrow \hat{\beta}^{S_{t-1}}_{\text{OLS}}(X, Y)$ ▷ Fit OLS on subset S_{t-1} from prev. iteration

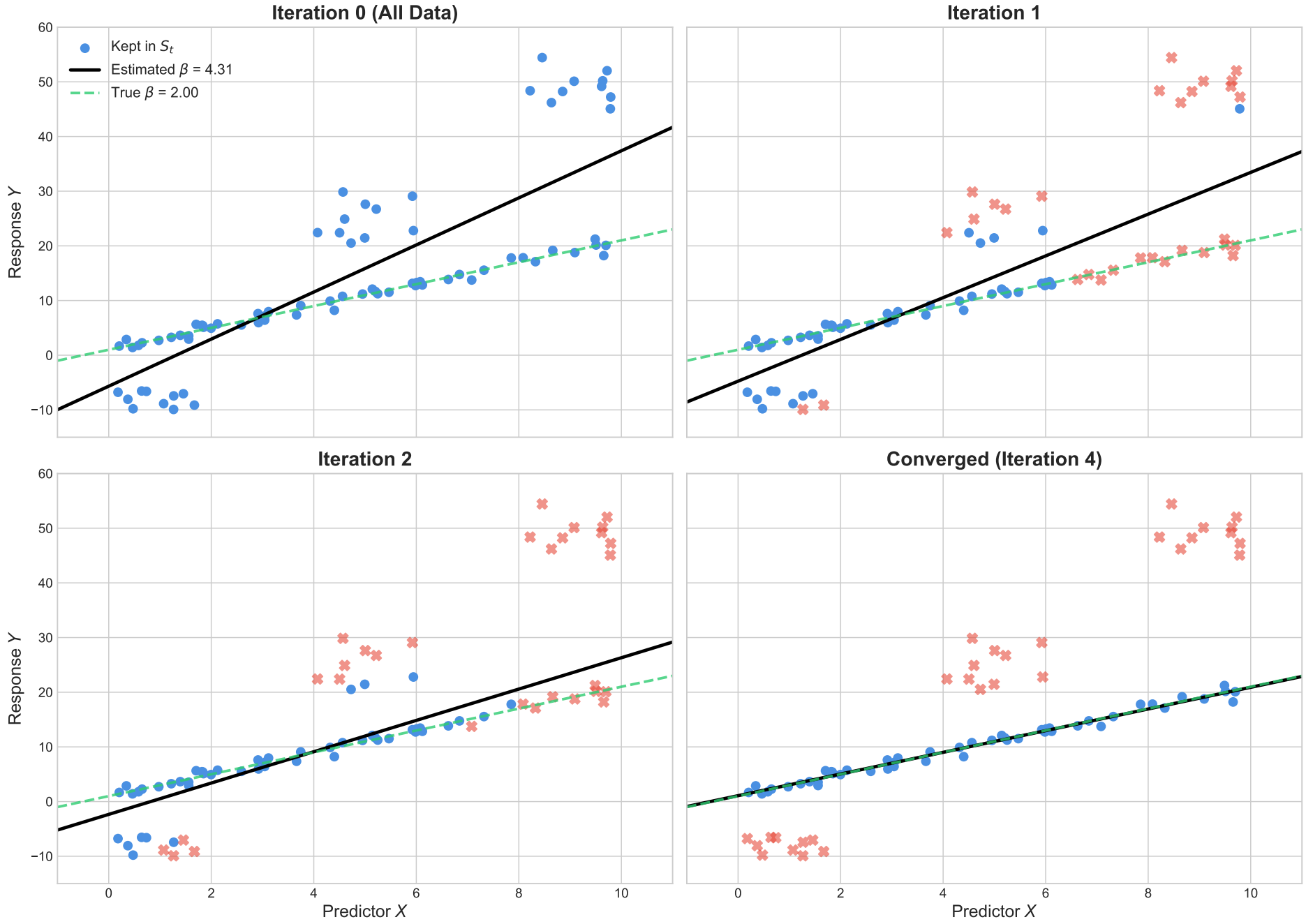
  $v \leftarrow |Y - X\hat{\beta}^t_{\text{Tor}}|$ ▷ Calc residuals for ALL data points

  $S_t \leftarrow \text{HT}(v, a)$ ▷ Keep a many points with smallest residuals

  $e \leftarrow |Y_{S_t} - X_{S_t}\hat{\beta}^t_{\text{Tor}}|$ ▷ Measure error on the new clean subset S_t

end while

return $\hat{\beta}^{n,a}_{\text{Tor}} := \hat{\beta}^t_{\text{Tor}}$


Iteration 0 (All Data)
Iteration 1
Iteration 2
Converged (Iteration 4)
Kept in S_t
Estimated β = 4.31
True β = 2.00
Response Y
Predictor X
60
50
40
30
20
10
0
−10
0
2
4
6
8
10

Empirical Convergence of Torrent

n	**mean**	**min**	**max**
10	2.42	2	3
100	5.14	3	8
1000	8.26	4	13

Convergence Speed

Even for a large sample size of $n = 1000$, Torrent converges in < 15 **iterations**.

Experimental setting: X and U are band-limited processes, theshold param. $a = 0.7$, elements of G_n chosen uniformly at random with probability 0.25 [3].

Outline

1. Introduction
2. Theoretical Setting
3. Methodology
4. Theoretical Guarantees
5. Experiments
6. Conclusion & Future Work

Setting the Stage for Proofs for Torrent

Key Assumptions

1. **Setting 2:** Let $n \geq d$. We observe $X \in \mathbb{R}^{n \times d}$ and $Y \in \mathbb{R}^n$ defined by:

$$Y := X\beta + \epsilon + o$$

   where $\epsilon \in \mathbb{R}^n$ is noise and $o \in \mathbb{R}^n$ are sparse outliers supported on G_n (i.e., $o_i = 0$ for $i \notin G_n$). We call G_n^c the **inliers**.
2. **Bounded Outliers G_n:** # confounded frequencies $|G_n| \leq c_n$.
3. **Mistakes-Stability-Ratio:** As $n \to \infty$, it holds with high probability that:

$$\eta := \max_{S \text{ s.t. } |S| = n - c_n} \frac{||X_{V(S)}||_2}{\sqrt{\lambda_{\min}(X_S^\top X_S)}} < \frac{1}{\sqrt{2}}$$

   where $S \subseteq \{1, ..., n\}$ and $V(S) = (S \cup G_n^c) \setminus (G_n^c \cap S)$ is the symmetric difference between S and G_n^c.

Intuitions:

- Should η better be nearly 0 or close to our upper bound?
- What happens if X_S is singular for some S?

Proofs - Torrent (deterministic)

Guarantee of Torrent (Theorem 3)

Let S_t be the final estimated subset of Torrent on data X, Y using threshold parameter $a_n := n - c_n$. Then

$$\|\hat{\beta}_{\text{Tor}}^{n,a_n} - \beta\|_2 \leq \text{Err}_{\text{OLS}} + \frac{\sqrt{2}\|X_{V(S_t)}\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|\epsilon_{V(S_t)}\|_2}{\sqrt{\lambda_{\min}(X_{S_t}^\top X_{S_t})}(1 - \sqrt{2}\eta)}.$$

with $\text{Err}_{\text{OLS}} = \|(X_{S_t}^\top X_{S_t})^{-1} X_{S_t}^\top \epsilon_{S_t}\|_2$ *being the baseline error of the clean subset.*

Intuitions:

- What happens in case of no outliers at all?
- What happens if $\eta \approx \frac{1}{\sqrt{2}}$ or X_{S_t} is singular?

Detailed Proof - Theorem 3

Assume $X_{S_{t-1}}^\top X_{S_{t-1}}$ is invertible.

1. Optimality Bound (Eq. 27)

We use the optimality of the model $\hat{\beta}_{\text{Tor}}^t$ on the active set S_{t-1}:

$$\begin{aligned}
\|\hat{\beta}_{\text{Tor}}^t - \beta\|_2 &= \left\|(X_{S_{t-1}}^\top X_{S_{t-1}})^{-1}(X_{S_{t-1}}^\top \epsilon_{S_{t-1}} + X_{S_{t-1}}^\top o_{S_{t-1}})\right\|_2 \\
&\leq \left\|(X_{S_{t-1}}^\top X_{S_{t-1}})^{-1} X_{S_{t-1}}^\top \epsilon_{S_{t-1}}\right\|_2 \\
&\quad + \left\|(X_{S_{t-1}}^\top X_{S_{t-1}})^{-1} X_{S_{t-1}}^\top\right\|_2 \|o_{S_{t-1}}\|_2
\end{aligned}$$

2. Hard Thresholding Guarantee (Eq. 28)

Following the HT-step and using relation $|G_n| \leq c_n \implies |G_n^c| \geq n - c_n$:

$$\begin{aligned}
\|X_{S_t}(\beta - \hat{\beta}_{\text{Tor}}^t) + \epsilon_{S_t} + o_{S_t}\|_2^2 &= \|Y_{S_t} - X_{S_t}\hat{\beta}_{\text{Tor}}^t\|_2^2 \\
&\leq \|Y_{G_n^c} - X_{G_n^c}\hat{\beta}_{\text{Tor}}^t\|_2^2 \\
&= \|X_{G_n^c}(\beta - \hat{\beta}_{\text{Tor}}^t) + \epsilon_{G_n^c}\|_2^2
\end{aligned}$$

Bounding the Outliers $\|o_{S_t}\|_2$:
Let $H_t := S_t \setminus G_n^c$ and $M_t := G_n^c \setminus S_t$. We know from above:

$$\|X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon_{H_t} + o_{H_t}\|_2 \leq \|X_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon_{M_t}\|_2$$

Let $H_t \cup M_t = V(S_t) =: V_t$. We can then expand by triangle inequality:

3. Outlier Expansion (Eq. 29)

$$\begin{aligned}
\|o_{S_t}\|_2 &\leq \|X_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|\epsilon_{H_t}\|_2 + \|\epsilon_{M_t}\|_2 \\
&\leq \sqrt{2}\,\|X_{V_t}\|_2\,\|\beta - \hat{\beta}^t_{\text{Tor}}\|_2 + \sqrt{2}\,\|\epsilon_{V_t}\|_2 \\
&\leq \sqrt{2}\,\|X_{V_t}\|_2 \Big(\Big\|(X_{S_{t-1}}^\top X_{S_{t-1}})^{-1} X_{S_{t-1}}^\top \epsilon_{S_{t-1}}\Big\|_2 \\
&\qquad + \Big\|(X_{S_{t-1}}^\top X_{S_{t-1}})^{-1} X_{S_{t-1}}^\top\Big\|_2 \|o_{S_{t-1}}\|_2 \Big) + \sqrt{2}\,\|\epsilon_{V_t}\|_2
\end{aligned}$$

- Add and subtract: $\|o_{S_t}\|_2 = \big\|(X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon_{H_t} + o_{H_t}) - (X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon_{H_t})\big\|_2$
- Apply $(|a| + |b|)^2 \leq 2(a^2 + b^2)$ twice to merge H_t and M_t into V_t:
  - For the noise: $(\|\epsilon_{H_t}\|_2, \|\epsilon_{M_t}\|_2) \to \sqrt{2}\|\epsilon_{V_t}\|_2$
  - For the signal: $(\|X_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2, \|X_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2) \to \sqrt{2}\|X_{V_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2$
- Observe $\|o_{H_t}\|_2 = \|o_{S_t}\|_2$, recalling $o_i = 0$ for $i \notin G_n$

Let t be the final iteration of Torrent, meaning $S_t = S_{t-1}$ (proof of this in the backup).
By assumption:

$$\eta = \max_{S \subseteq \{1,\ldots,n\} \text{ s.t. } |S| = a_n} \frac{\|X_{V(S)}\|_2}{\sqrt{\lambda_{\min}(X_S^\top X_S)}} = \max_S \|X_{V(S)}\|_2 \|(X_S^\top X_S)^{-1} X_S^\top\|_2 < \frac{1}{\sqrt{2}}$$

- Recall the matrix norm property: $\|A\|_2 = \sqrt{\lambda_{\max}(A^\top A)} = \sqrt{\lambda_{\max}(AA^\top)}$.
- Set $A := (X_S^\top X_S)^{-1} X_S^\top$ and observe $AA^\top = (X_S^\top X_S)^{-1}$
- $\|(X_S^\top X_S)^{-1} X_S^\top\|_2 = \|A\|_2 = \sqrt{\lambda_{\max}(AA^\top)} = \sqrt{\lambda_{\max}((X_s^\top X_s)^{-1})} = 1/\sqrt{\lambda_{\min}((X_s^\top X_s))}$

4. Applying η to isolate o_{S_t} (Eq. 30)

Distributing terms in Eq. 29 and substituting the η bound:

$$\|o_{S_t}\|_2 \leq \underbrace{\sqrt{2}\|X_{V_t}\|_2 \left\| \left(X_{S_t}^\top X_{S_t}\right)^{-1} X_{S_t}^\top \right\|_2}_{\leq \sqrt{2}\eta} \|o_{S_t}\|_2$$

$$+ \sqrt{2}\|X_{V_t}\|_2 \left\| \left(X_{S_t}^\top X_{S_t}\right)^{-1} X_{S_t}^\top \epsilon_{S_t} \right\|_2 + \sqrt{2}\|\epsilon_{V_t}\|_2$$

$$\implies \|o_{S_t}\|_2 \leq \frac{\sqrt{2}\|X_{V_t}\|_2 \cdot \text{Error}_{\text{OLS}} + \sqrt{2}\|\epsilon_{V_t}\|_2}{1 - \sqrt{2}\eta}.$$

5. Putting everything together

Using $S_t = S_{t-1}$, Eq. 27, Eq. 30:

$$\begin{aligned}
\|\hat{\beta}^t_{\text{Tor}} - \beta\|_2 &\leq \left\|(X^\top_{S_{t-1}} X_{S_{t-1}})^{-1} X^\top_{S_{t-1}} \epsilon_{S_{t-1}}\right\|_2 + \left\|(X^\top_{S_{t-1}} X_{S_{t-1}})^{-1} X^\top_{S_{t-1}}\right\|_2 \|o_{S_{t-1}}\|_2 \\
&= \text{Err}_{\text{OLS}} + \left\|(X^\top_{S_t} X_{S_t})^{-1} X^\top_{S_t}\right\|_2 \cdot \|o_{S_t}\|_2 \\
&= \text{Err}_{\text{OLS}} + \frac{1}{\sqrt{\lambda_{\min}((X_s^\top X_s))}} \cdot \|o_{S_t}\|_2 \\
&\leq \text{Err}_{\text{OLS}} + \frac{\sqrt{2}\|X_{V_t}\|_2 \cdot \text{Error}_{\text{OLS}} + \sqrt{2}\|\epsilon_{V_t}\|_2}{\sqrt{\lambda_{\min}(X^\top_{S_t} X_{S_t})}(1 - \sqrt{2}\eta)}
\end{aligned}$$

Remark: This deterministic bound can be extended to a probabilistic bound under some Assumptions of noise, i.e.

Guarantee of Torrent with sub-Gaussian noise (Corollary 4)

If the rows of X are i.i.d. standard Gaussian random vectors (i.e. $\lambda_{a_n}(X) \in \Omega_{\mathbb{P}}(\sqrt{n})$ [1]) and $c_n \in o(n/\log(n))$, then $\hat{\beta}^{n,a_n}_{\text{Tor}}$ is consistent.

Convergence properties of DecoR-Tor (Theorem 6)

Let c_n be a known sequence of non-negative natural numbers such that $|G_n| \leq c_n$ and let Assumption 2 in [1] be satisfied. Assume that for all $\delta > 0$ there exists $\bar{n} \in \mathbb{N}$ such that for all $n \geq \bar{n}$ it holds that

$$\mathbb{P}\left[\max_{S \subseteq \{1,\ldots,n\} \text{ s.t. } |S| = n - c_n} \frac{\|(X_\phi^n)_{V(S)}\|_2}{\sqrt{\lambda_{\min}\left((X_\phi^n)_S^\top (X_\phi^n)_S\right)}} < \frac{1}{\sqrt{2}}\right] \geq 1 - \delta.$$

If DecoR is executed with Torrent and the sequence of threshold parameters $a_n = n - c_n$, then

$$\|\hat{\beta}_{\text{DecoR}}^{\phi,n} - \beta\|_2 \in \mathcal{O}_{\mathbb{P}}\left(\sigma\sqrt{\frac{c_n \log(n/c_n)}{n}}\right).$$

Insights:

- $\hat{\beta}_{\text{DecoR}}^{\phi,n}$ is a consistent estimator of β, if $\sigma = 0$ or $c_n \in o(n)$
- if noise is non-zero and c_n is asymptotically proportional to n, then the estimation error is bounded by σ, up to constant factors.

Outline

1. Introduction
2. Theoretical Setting
3. Methodology
4. Theoretical Guarantees
5. Experiments
6. Conclusion & Future Work

Synthetic Data - Data Generation

Recall:

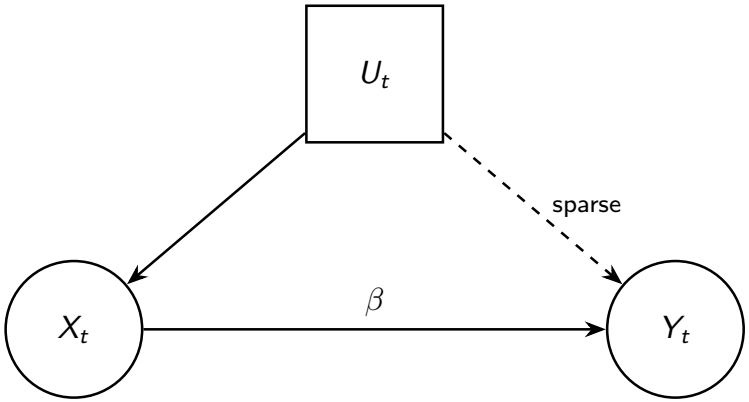


Figure: Causal Graph of X and Y with hidden confounder U [4]

We set in our one-dimensional scenario:

- $U := \text{AR}(1) + \epsilon_u$ with AR(1) := Ornstein-Uhlenbeck process and noise $\epsilon_u \sim N(0, 1)$
- $U_{sparse} = \text{sparsify}(U, |G_n| = 0.25 \cdot n, \text{cosine-basis})$
- $X := \text{AR}(2) + U_{sparse} + \epsilon_x$ with $\epsilon_x \sim N(0, 1)$
- $Y := 3X + 10 \cdot U_{spasre} + \epsilon_y$ with $\epsilon_y \sim N(0, 1)$

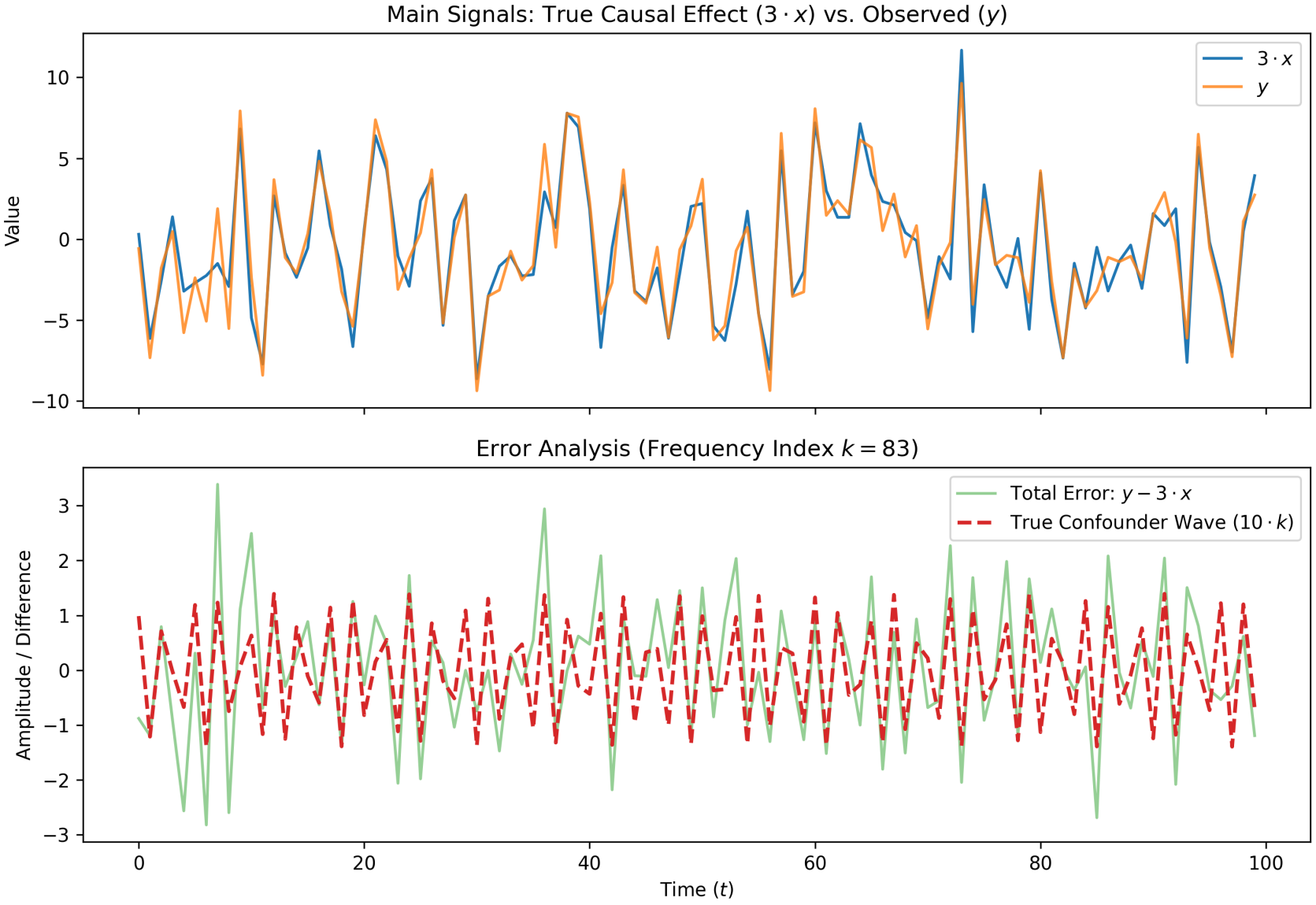


Figure: Visualization of the true causal signal $(3 \cdot X)$, observed data (Y), and the isolated sparse confounder (U_{sparse}) with $n = 100$, $G_n = \{k\}$

Synthetic Data - Experimental Results

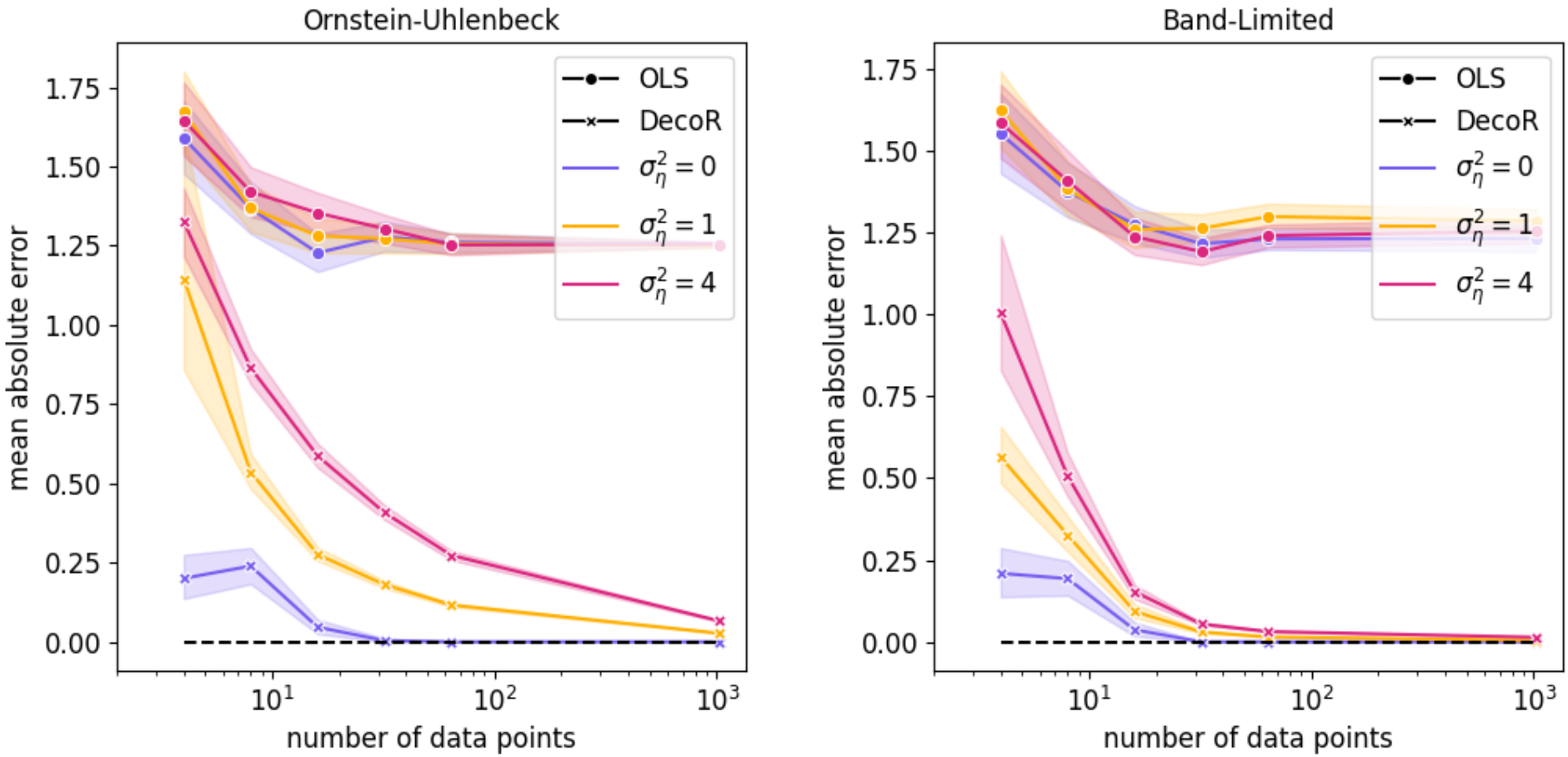
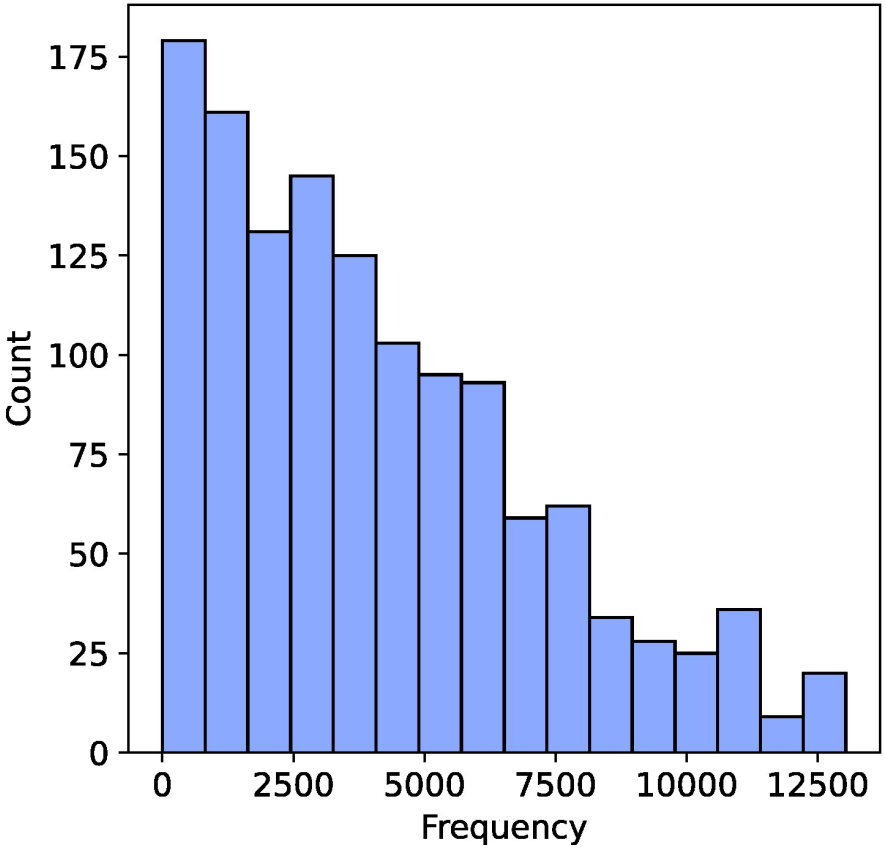
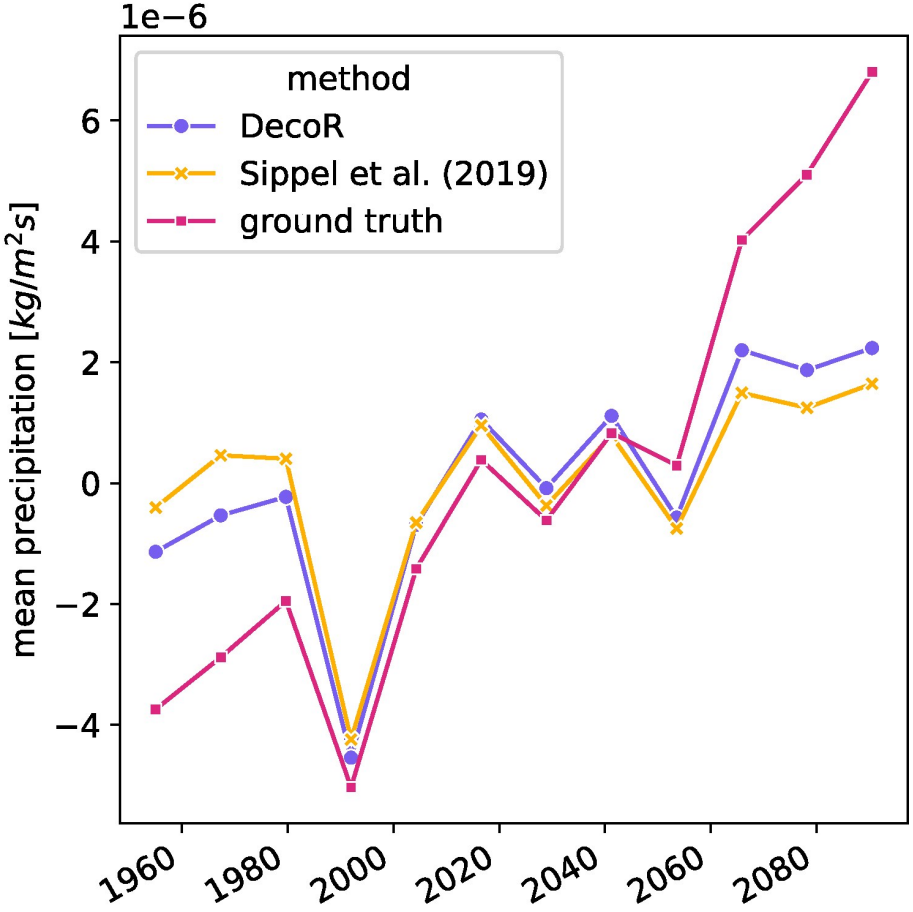


Figure: Comparison of DecoR performance against OLS with $|G_n| = 0.25$ and cosine-basis. In the right figure, band B of size 50 is used.

Application: Earth System Science





Findings: DeCoR automatically identifies and filters out the slow, low-frequency components associated with climate change, accurately isolating the short-term "weather" signal β.

Outline

1. Introduction
2. Theoretical Setting
3. Methodology
4. Theoretical Guarantees
5. Experiments
6. Conclusion & Future Work

Conclusion

- **The Problem:** Unobserved confounders severely bias standard regression in time series.
- **The DeCoR Insight:** By moving to the frequency domain, we exploit the sparsity of confounders.
- **The Method:** Treating confounded frequencies as adversarial outliers allows robust regression (like Torrent) to recover the true causal effect.

Future Work

Beyond the "Dense Y" extension, the authors identify several promising directions for future research:

1. **Irregular Intervals:** Adapting the spectral transformations for time series data that is not sampled uniformly.
2. **General Noise:** Extending the theoretical consistency proofs to handle heteroskedastic (non-constant variance) noise distributions.
3. **Confidence Intervals:** Developing provably valid confidence intervals rather than relying solely on heuristic methods.

References I

Kush Bhatia, Prateek Jain, and Purushottam Kar.
Robust regression via hard thresholding, 2015.

Xin Lan and Ralph Keeling.
Trends in atmospheric carbon dioxide, 2026.
Additional data accessible via Scripps Institution of Oceanography: https://scrippsco2.ucsd.edu/.

Felix Schur and Jonas Peters.
Decor: Deconfounding time series with robust regression, 2024.

Sebastian Sippel, Nicolai Meinshausen, Anna Merrifield, Flavio Lehner, Angeline G. Pendergrass, Erich Fischer, and Reto Knutti.
Uncovering the forced climate response from a single ensemble member using statistical learning.
Journal of Climate, 32(17):5677 – 5699, 2019.

References II

Patrick Zippenfenig.
Open-Meteo.com Weather API.

Backup

Proofs - Torrent (probabilistic)

Define the square root of minimum eigenvalue of all possible clean subsets as

$$\lambda_{a_n}(X) := \min_{S \subseteq \{1,\ldots,n\} \text{ s.t. } |S| = a_n} \sqrt{\lambda_{\min}(X_S^\top X_S)}$$

Guarantee of Torrent with sub-Gaussian noise (Corollary 4)

Let ϵ be i.i.d. zero-mean sub-Gaussian noise with variance proxy σ^2 and assume $c_n < n/2$. Then, there exists a constant $K > 0$ s.t. $\forall\ \delta > 0$ with probability $\geq 1 - \delta$:

$$\|\hat{\beta}_{\text{Tor}}^{n,a_n} - \beta\|_2 \leq \frac{\sigma}{\lambda_{a_n}(X)} \left(1 + \frac{1}{1 - \sqrt{2}\eta}\right) \left(\sqrt{d} + \sqrt{2c_n} L_{n,\delta}\right) + \frac{2\sigma\sqrt{c_n}\,(1 + L_{n,\delta})}{\lambda_{a_n}(X)(1 - \sqrt{2}\eta)}.$$

with $L_{n,\delta} := \sqrt{K \log(2en/c_n\delta)}$ representing the logarithmic search penalty.

Insights:

- What happens if $c_n = 0$? Compare it to standard OLS.
- If the rows of X are i.i.d. standard Gaussian random vectors (i.e. $\lambda_{a_n}(X) \in \Omega_{\mathbb{P}}(\sqrt{n})$ [1]) and $c_n \in o(n/\log(n))$, then $\hat{\beta}_{\text{Tor}}^{n,a_n}$ is consistent.

Convergence properties of DecoR-Tor (Theorem 6) - shortened

$$\|\hat{\beta}^{\phi,n}_{\text{DecoR}} - \beta\|_2 \in \mathcal{O}_{\mathbb{P}}\left(\sigma\sqrt{\frac{c_n \log(n/c_n)}{n}}\right).$$

Proof ideas:

- By some assumptions, note that $\sqrt{\lambda_{\min}\left((X^n_\phi)^\top_S (X^n_\phi)_S\right)} \geq c'$ with high probability.
- each component of $\eta^n_\phi = \{T^{\phi,n}_k(\eta)\}_{k\leq n} = \frac{1}{n}\sum_{l=1}^n (\eta_{Tl/n})_k \cdot \phi_k(Tl/n)$ has (by factor n) reduced variance $\bar{\sigma}^2 = \sigma^2/n$
- Use **Corollary 4** and derive Big-$\mathcal{O}$ result:

$$\|\hat{\beta}^{n,a_n}_{\text{Tor}} - \beta\|_2 \leq \frac{\sigma}{\lambda_{a_n}(X)}\left(1 + \frac{1}{1-\sqrt{2}\eta}\right)\left(\sqrt{d} + \sqrt{2c_n K \log\left(2en/c_n\delta\right)}\right) + \frac{2\sigma\sqrt{c_n}\left(1 + \sqrt{K\log\left(2en/c_n\delta\right)}\right)}{\lambda_{a_n}(X)(1-\sqrt{2}\eta)}.$$